Biopsychosocial and Epigenetic Markers of Heat-Stress and Posttraumatic Outcomes in Emergency Responders

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Firefighters showed a greater decline in Hannum epigenetic age acceleration than paramedics between assessments.

-2.33 years
Difference in change relative to paramedics

01 Study and analysis

Emergency responder professions differ in their physical and psychosocial exposures. We examined whether epigenetic ageing trajectories differed between professions and explored methylation-associated biological processes using a heat-response gene reference set.

332 **participants at baseline**

- **168** Paramedics
- **99** Emergency medical dispatchers
- **65** Firefighters



Figure 1. Three emergency responder professions were assessed across two timepoints.

Psychosocial assessments and saliva-derived DNA methylation were examined across two timepoints approximately 12 months apart. Methylation was measured using Illumina MethylationEPIC v2 arrays.

Epigenetic age acceleration was assessed using Horvath, Hannum [1], PhenoAge and GrimAge clocks. Longitudinal models included participant random intercepts and adjusted for sex, ethnicity, trauma exposure, BMI and estimated cell composition.

02 Hannum change over time

The Timepoint × Profession interaction tested whether changes from T1 to T2 differed by profession, using paramedics as the reference group.

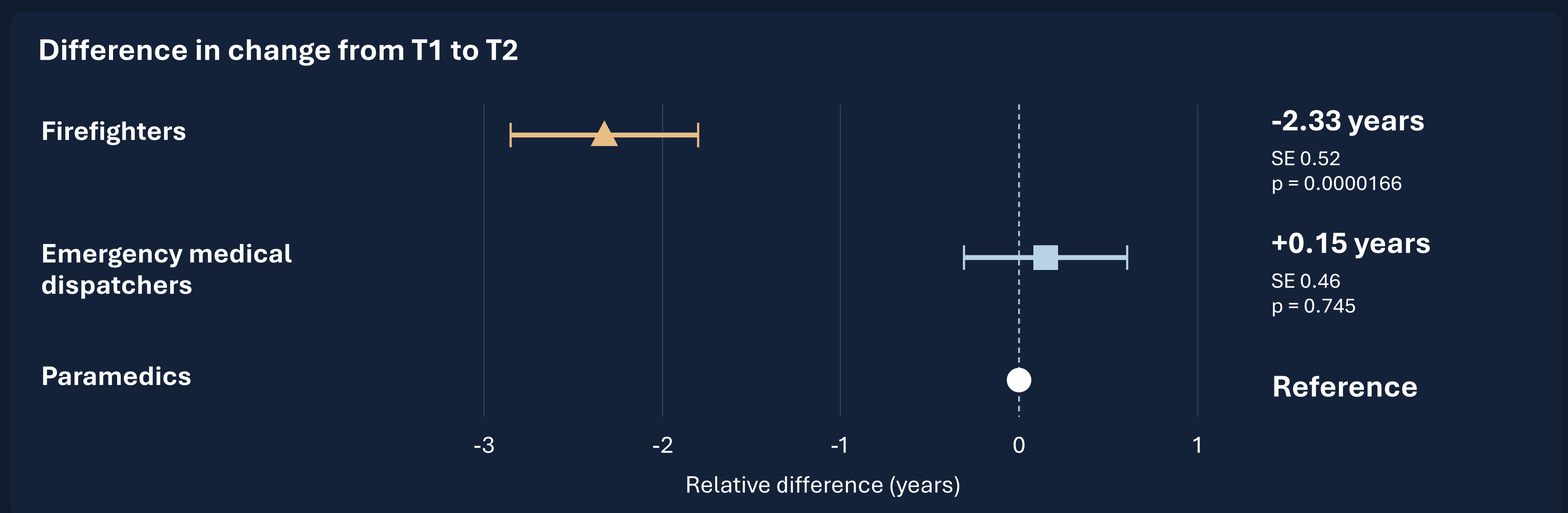


Figure 2. Points show how the change in Hannum age acceleration from T1 to T2 differed from the change in paramedics; bars show ±1 standard error. Estimates are the Timepoint × Profession coefficients from a covariate-adjusted mixed-effects model with participant random intercepts: 464 observations from 330 participants. P-values are nominal tests of the individual interaction coefficients.

Among the four clocks, Hannum showed evidence of profession-related differences in change over time (Timepoint × Profession). Firefighters showed a greater decline in Hannum age acceleration between T1 and T2 than paramedics. The change among emergency medical dispatchers did not differ significantly from paramedics.

At baseline, firefighters had higher PhenoAge acceleration than paramedics: β = 1.52 years, SE = 0.76, nominal p = 0.047.

03 Biological processes

GOstats examined genes linked to the Timepoint × Profession interaction, comparing DNA methylation changes over time in firefighters with those in paramedics. Leading terms concerned morphogenesis, actin filament-based processes and angiogenesis.

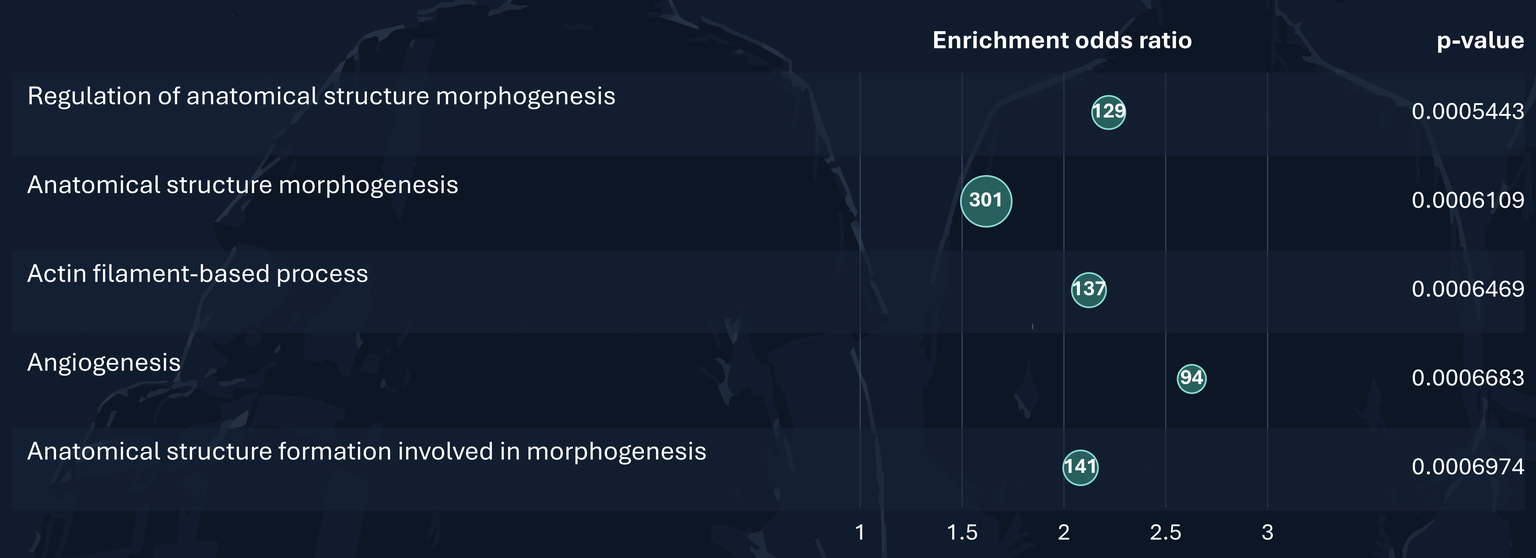


Figure 3. Five highest-ranked GO biological processes for the Timepoint × Profession interaction, comparing methylation changes over time in firefighters with paramedics. Position shows enrichment odds ratio; bubble area and numbers show selected gene counts. GOstats used CpGs selected at nominal p < 0.05 and a heat-response reference universe.

In separate GOmeth analyses of time and the omnibus Timepoint × Profession interaction, no GO or KEGG terms met FDR < 0.05 [2].

04 Interpretation

Possible immune-related explanation

The decline could reflect changes in immune-related processes, including recovery from acute immune activation. Firefighting involves exposures such as heat, smoke, disrupted sleep and physical exertion. If T1 coincided with recent exposure and T2 with relative recovery, this could contribute to the observed pattern.

Blood-based research shows that Hannum estimates are sensitive to immune cell composition. Immune recovery therefore remains a possible explanation, rather than an established mechanism in the study [4].

Psychological symptoms and positive factors

Psychological symptoms and positive psychological factors showed no association across the four epigenetic clocks. Symptoms were assessed approximately 12 months apart; a relatively short interval over which changes in DNAm-based ageing may be subtle, potentially limiting detection of associations. Technical variability can also complicate the interpretation of small longitudinal changes [3].

Next step

Longer follow-up combining exposure measurements, inflammatory markers and repeated DNA methylation assessments could help distinguish transient occupational responses from sustained changes in epigenetic ageing.

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[1] Hannum G et al. Molecular Cell. 2013;49:359-367. [2] Maksimovic J et al. Genome Biology. 2021;22:173.
[3] Higgins-Chen AT et al. Nature Aging. 2022;2:644-661. [4] Zhang Z et al. Aging Cell. 2024;23:e14071.

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